Year 13 Further Maths - Lines and planes in space

Similar questions - Tetrahedra, volume and rotation about an axis

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The points P, Q and R have coordinates $(1,0,2)$, $(3,1,0)$ and $(2,3,1)$.

(a) Show that the acute angle θ between PQ and PR satisfies $\sin\theta = \dfrac{5\sqrt{22}}{33}$. **[3]**

(b) Find an equation of the plane Π containing P, Q and R in the form $ax+by+cz=d$ with integers a,b,c,d. **[3]**

(c) The point S has coordinates $(4,2,5)$. By finding the shortest distance from S to Π, show that the volume of the tetrahedron $PQRS$ is 5. **[4]**

(d) The tetrahedron is rotated about the y-axis through a non-zero angle ϕ to $P'Q'R'S'$, where S' has x-coordinate 4. Using the matrix $\begin{pmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{pmatrix}$, determine the exact coordinates of R'. **[5]**

2 The point A has coordinates $(2,-1,4)$ and the plane Π has equation $x+2y-2z=3$.

(a) Find the shortest distance from A to Π. **[3]**

(b) Find the coordinates of the point on Π closest to A. **[3]**

Question 1

(a) $\overrightarrow{PQ} = (2,1,-2)$, $|PQ| = 3$; $\overrightarrow{PR} = (1,3,-1)$, $|PR| = \sqrt{11}$. $|\overrightarrow{PQ} \times \overrightarrow{PR}| = |(5,0,5)| = 5\sqrt{2}$. $\sin\theta = \dfrac{5\sqrt{2}}{3\sqrt{11}} = \dfrac{5\sqrt{22}}{33}$

(b) Normal $(5,0,5) \propto (1,0,1)$, so $x+z=d$ with P: $1+2=3$. $x+z=3$ (check Q: $3+0$, R: $2+1$)

(c) S: $4+5=9$, distance $= \dfrac{|9-3|}{\sqrt{2}} = 3\sqrt{2}$. Area of $PQR = \frac{1}{2} \times 5\sqrt{2}$. Volume $= \frac{1}{3} \times \frac{5\sqrt{2}}{2} \times 3\sqrt{2} = 5$

(d) $x'_S = 4\cos\phi + 5\sin\phi = 4$ gives $5\sin\phi = 4(1-\cos\phi)$, so $\tan\frac{\phi}{2} = \frac{5}{4}$ (for $\phi \neq 0$). Then $\cos\phi = \dfrac{1-25/16}{1+25/16} = -\dfrac{9}{41}$ and $\sin\phi = \dfrac{2\cdot 5/4}{1+25/16} = \dfrac{40}{41}$. $R' = (2\cos\phi + \sin\phi,\ 3,\ -2\sin\phi + \cos\phi) = \left(\dfrac{22}{41}, 3, -\dfrac{89}{41}\right)$

Question 2

(a) $\dfrac{|2-2-8-3|}{\sqrt{1+4+4}} = \dfrac{11}{3}$

(b) The foot is $A - t\mathbf{n}$ with $\mathbf{n} = (1,2,-2)$: $(2-t) + 2(-1-2t) - 2(4+2t) = 3$ gives $-8-9t = 3$, $t = -\frac{11}{9}$. Foot $= A + \frac{11}{9}\mathbf{n} = \left(\frac{29}{9}, \frac{13}{9}, \frac{14}{9}\right)$